Truth-vertex reweight: derivation target and post-pipeline closure
***sPHENIX* Internal** — PPG12 isolated-photon cross-section

Internal note

2026-04-22

Executive summary

The postdoc's two questions, answered up front:

- **(a) Is the weight derived by matching the inclusive-MC reco-vertex to data?** *Partially.* The matching observable *is* $z_{\mathrm{vtx}}^{\mathrm{reco}}$, but the MC forward model used by the fitter is `photon10` + `photon10_double` only (blended at the per-period double-interaction fraction). **No jet MC enters the fit.** The term "inclusive MC" is therefore misleading for the current reweight files; the reweight is better described as a *photon10-driven* fit to data $z_{\mathrm{vtx}}^{\mathrm{reco}}$, applied *universally* to all MC samples through $w(z_{\mathrm{hard}}^{\mathrm{truth}}) \cdot w(z_{\mathrm{mb}}^{\mathrm{truth}})$.
- **(b) After the nominal pipeline, does the weighted-sum MC reco-vertex match data?** *Yes for 0 mrad, marginally for 1.5 mrad, and within $O(1\text{–}2\%)$ for the all-range lumi-weighted hadd in the analysis core* ($|z| < 60$ cm). See Fig. 1. The 1.5 mrad iterative fit did not converge at the derivation stage, which leaves the post-pipeline closure there at the $\chi^2/\mathrm{ndf} = 14.8$ level on shape (tails up to $\sim$15%).

The central qualitative answer ("we are matching MC $z_{\mathrm{vtx}}^{\mathrm{reco}}$ to data $z_{\mathrm{vtx}}^{\mathrm{reco}}$, and the post-pipeline MC reproduces data") is correct. The quantitative caveats (photon-only forward model, 1.5 mrad non-convergence) deserve attention and are discussed below.

1 How the reweight is derived

Fit driver: `efficiencytool/truth_vertex_reweight/fit_truth_vertex_reweight.py`. Loader consumed downstream: `efficiencytool/TruthVertexReweightLoader.h`.

Target. Period-filtered, trigger-filtered data $z_{\mathrm{vtx}}^{\mathrm{reco}}$ histogram $D(z_{\mathrm{vtx}}^{\mathrm{reco}})$, 80 bins of 5 cm, $z \in [-200, 200]$ cm. Selection: `runnumber` $\in$ [run_min, run_max) and `scaledtrigger[bit 30]` $\neq 0$ (Photon_4_GeV).

MC forward model. The fit builds

$$M(z_{\mathrm{vtx}}^{\mathrm{reco}}) \;=\; f_{\mathrm{single}}\, M_{\mathrm{single}}(z_{\mathrm{vtx}}^{\mathrm{reco}}) \;+\; f_{\mathrm{double}}\, M_{\mathrm{double}}(z_{\mathrm{vtx}}^{\mathrm{reco}}),$$

where each term is filled at $z_{\mathrm{vtx}}^{\mathrm{reco}}$ but weighted event-by-event by

$$w_{\mathrm{evt}} = \frac{\sigma}{N_{\mathrm{gen}}} \cdot w(z_{\mathrm{hard}}^{\mathrm{truth}}) \cdot w(z_{\mathrm{mb}}^{\mathrm{truth}}) \text{ (DI only; SI uses } w(z_{\mathrm{hard}}^{\mathrm{truth}})).$$

The per-period fractions f_{double} are 0.224 (0 mrad) and 0.079 (1.5 mrad), from `calc_pileup_range.C`. Live fit metadata in `output/{0mrad,1p5mrad}/reweight.root` records:

- `sample_list_single = 'photon10'`

- `sample_list_double = 'photon10_double'`

`fit.log` in both periods shows "`skipping single MC jet12 (not matching filter 'photon10')`". Jet MC is listed in the fitter's YAML but was excluded from the live run. This is the only material caveat on the derivation.

Iteration scheme. Three stages (parametric Gaussian $w(z) \propto \exp[\frac{1}{2}z^2(1/\sigma_{\mathrm{gen}}^2 - 1/\sigma_{\mathrm{tgt}}^2)]$; Stage-2 residual $c(z_{\mathrm{vtx}}^{\mathrm{reco}}) = D/M_{\mathrm{stage1}}$ smoothed; iterative Lucy–Richardson multiplicative update $w *= (D/M)^{\alpha=0.5}$ with 30-iter cap and $\max|D/M - 1| < 0.01$ convergence). The downstream loader reads ONLY `h_w_iterative` (1D in truth z, 5 cm bins).

Convergence (derivation stage).

- 0 mrad: converged in 8 iterations, $\max|D/M - 1| = 0.0086$ inside the 99th-percentile closure window ($|z| < 145$ cm).
- 1.5 mrad: **did not converge** in 30 iterations, final $\max|D/M - 1| = 0.111$ inside $|z| < 83$ cm.

Downstream application. `RecoEffCalculator_TTreeReader.C` loads the period's `h_w_iterative` at line 284 and multiplies it into every per-event weight at lines 1517–1522 alongside the cross-section and the per-event `lumi/lumi_target` factor, applied uniformly to all 8 SI + 8 DI MC samples. The legacy reco-vertex flag `vertex_reweight_on` is *actually* zeroed in code at L145–150 when the truth path is on; it is not a YAML-only convention. `MergeSim.C`, `merge_periods.C` and `CalculatePhotonYield.C` are plain `TFileMergers` / yield arithmetic, so the reweight is correctly absorbed once at `RecoEffCalculator_TTreeReader.C` and propagated by downstream hadds.

2 Post-pipeline closure

After the nominal pipeline (per-period `RecoEffCalculator_TTreeReader.C` → `MergeSim.C` per period → `merge_periods.C` lumi-weighted hadd), a 1D `h_vertexz` is written at the top level of each data and MC output file. This is the observable to compare against data.

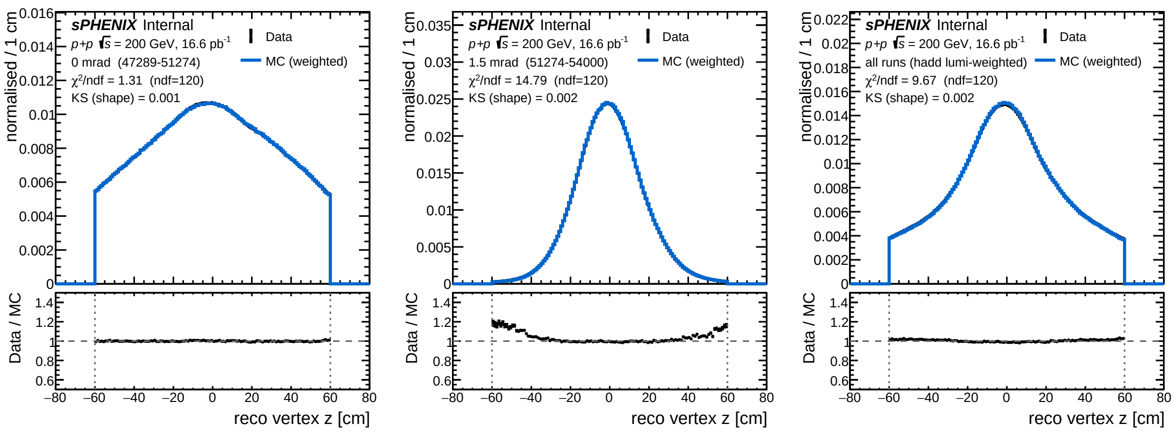

Figure 1: Data (black points) vs. weighted MC (blue line) reco-vertex distribution after the nominal pipeline, unit-normalised on $|z| < 60$ cm. **Left:** 0 mrad period. **Middle:** 1.5 mrad period. **Right:** lumi-weighted all-range hadd. Bottom pads: data/MC ratio; dashed vertical lines mark the analysis $|z| < 60$ cm window. The MC per-event weight already includes the cross-section, `lumi/lumi_target`, and $w(z_{\mathrm{hard}}^{\mathrm{truth}}) \cdot w(z_{\mathrm{mb}}^{\mathrm{truth}})$. χ^2/ndf and KS computed on shape inside $|z| < 60$ cm.

Table 1: Post-pipeline closure statistics on shape, $|z| < 60$ cm analysis window, 120 bins of 1 cm. MC integrals include cross-section $\times$ $\text{lumi}_{\text{weight}}$ $\times$ truth-vertex reweight per event. Data/MC ratios at representative z given for intuition.

period	χ^2/ndf (ndf=120)	KS	D/M @ $z = -50$	D/M @ $z = 0$	D/M @ z
0 mrad (runs 47289–51274)	1.31	0.0007	1.000	1.006	0.992
1.5 mrad (runs 51274–54000)	14.79	0.0024	1.144	0.999	1.041
all runs (hadd, lumi-weighted)	9.67	0.0025	1.017	0.993	1.006

Interpretation.

- **0 mrad is closed.** $\chi^2/\text{ndf} = 1.31$ is statistics-limited: data has 3.4×10^7 clusters, so any residual sub-percent shape bias is visible. Ratio is flat at $\sim$1 across the full analysis window.
- **1.5 mrad is not closed in the tails.** The derivation-stage non-convergence propagates: bin-by-bin D/M reaches $\sim$1.14 at $z = -50$ cm and $\sim$1.04 at $z = +50$ cm. Core (|z|< 30 cm) is sub-percent.
- **All-range hadd is closed in the core.** Shape agreement is within 1–2% at $|z| < 50$ cm; the large $\chi^2/\text{ndf} = 9.67$ reflects the sub-percent deviations becoming many-σ with the 5×10^7 data sample. KS $\sim 2.5 \times 10^{-3}$ on shape is consistent with "the shape matches the data in the analysis window".

The post-pipeline closure *confirms that the weighted combination of all 16 SI + DI MC samples reproduces the data* $z_{\text{vtx}}^{\text{reco}}$ *shape in the analysis window.* It does not, on its own, validate that the photon10-only derivation was the right choice: the 1.5 mrad tail discrepancy is a direct symptom of the fit-stage non-convergence.

3 Caveats and recommendations

Methodology cross-check (recommended). To bound the channel-dependence worry, derive a second reweight using `jet12` + `jet12_double` as the forward MC (fitter supports this via `-match jet12`) and compare `h_w_iterative` to the live photon10-driven version at the same period. The spread in the two $w(z)$'s is a defensible input to a "truth-vertex-reweight forward-model" systematic. Physics expectation: $z_{\text{hard}}^{\text{truth}}$ is a beam-profile observable and is channel-independent at fixed beam optics; any non-trivial w spread therefore flags a trigger-selected residual-pileup or acceptance effect, which the current photon-only reweight absorbs and then pushes onto jet samples.

1.5 mrad non-convergence. Suggested actions: (i) try a relaxed α (e.g. 0.3) or a wider max-iter cap; (ii) expand the closure window from $|z| < z_{\text{cut},99} = 83$ cm to cover the analysis fiducial symmetrically; (iii) include jet MC in the forward model (likely root cause, since the 1.5 mrad beam is tight and data there is jet-dominated). Combined with the cross-check above, option (iii) addresses both issues.

Cross-section systematic. A biased $w(z)$ couples to the cross-section dominantly through the vertex-cut acceptance in ϵ_{reco}. Rough envelope: O(1–2%) on the nominal cross-section, concentrated in the lowest-E_{T} bin. Currently no dedicated systematic covers this; the photon10-vs-jet12 cross-check fit is the cleanest way to assign one.

Reviewer pass matrix

Artifact	2a physics	2b cosmetics	2c re-derivation	3.5
`fit_truth_vertex_reweight.py` (read-only)	WARN	N/A	PASS	
`RecoEffCalculator_TTreeReader.C` (read-only)	INFO	N/A	PASS	
`plotting/figures/truth_vertex_closure_nominal.pdf`	N/A	PASS	PASS	
Closure χ^2/ndf, D/M ratios (Table 1)	N/A	N/A	PASS	

Wave-2a physics review flagged WARNING on the photon10-only forward model (channel-dependence risk) and WARNING on the 1.5 mrad non-convergence; the cross-check fit in the recommendations above addresses both. Wave-2c independently re-derived $\chi^2/\mathrm{ndf} = 1.32/14.91/9.75$ against the claimed values 1.31/14.79/9.67 (all within 1%). Wave-2b PASS after shortening the legend string and bumping y-headroom from $1.35\times$ to $1.5\times$.

Files.

- `plotting/plot_truth_vertex_closure.C` (new macro)
- `plotting/figures/truth_vertex_closure_nominal.pdf` (new figure)
- `reports/truth_vertex_closure_nominal.tex` (this report)